

B.Tech. II Sem.2024

Physics-PH-113

Quantum Mechanics

Matter Waves and Postulates of Wave Mechanics

Module-4: Quantum Mechanics: Planck's theory of black-body radiation, Compton effect, Wave particle duality, **De Broglie waves, Davisson and Germer's experiment, Uncertainty principle, Brief idea of Wave Packet, Wave Function and its physical interpretation, Schrodinger equation in one-dimension, free particle, particle in an infinite square well.**

Delivered by

Dr. Dheeraj Bhardwaj

Department of Physics

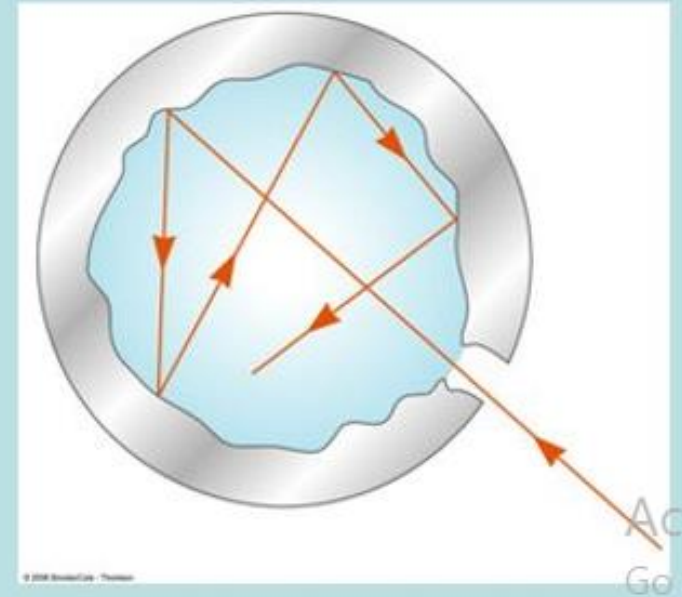
Blackbody Radiation and Planck's Hypothesis

❑ Thermal radiation

- An object at any temperature emits electromagnetic radiation called **thermal radiation**.
- The spectrum of the radiation depends on the temperature and properties of the object.
- From a classical point of view, thermal radiation originates from accelerated charged particles near the surface of an object.

❑ Blackbody

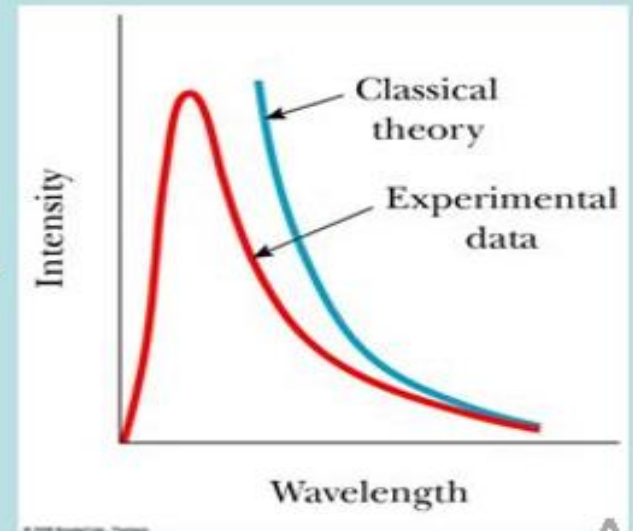
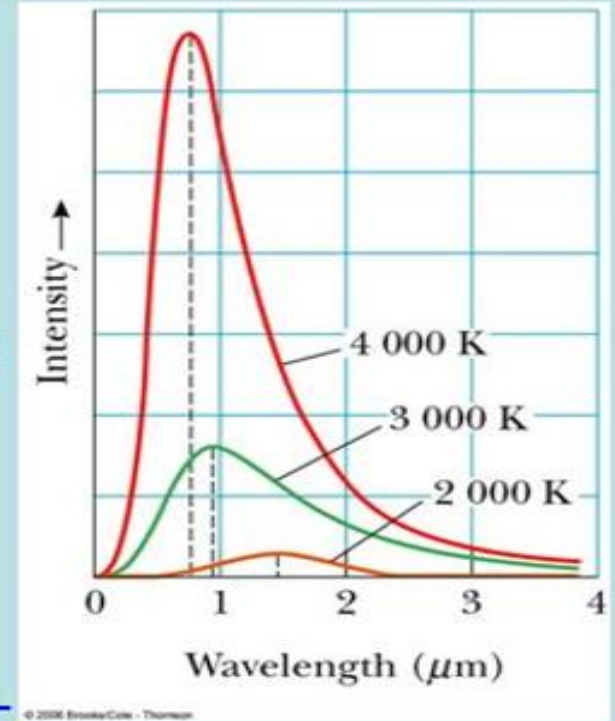
- Is an ideal system that absorbs all radiation incident on it.
- An opening in the cavity of a body is a good approximation of a blackbody.



Blackbody Radiation and Planck's Hypothesis

❑ Blackbody radiation

- The nature of the blackbody radiation depends only on the temperature of the body, not on the material composition of the object.
- The distribution of energy in blackbody radiation varies with wavelength and temperature.
 - The total amount of energy (area under the curve) it emits increase with the temperature.
 - The peak of the distribution shifts to shorter wavelengths. This shift obeys Wien's displacement law: $\lambda_{\text{max}} T = 0.2898 \times 10^{-2} \text{ m} \cdot \text{K}$
- The classical theory of thermal radiation at the end of 19th century failed to explain the distribution of energy of the blackbody radiation.

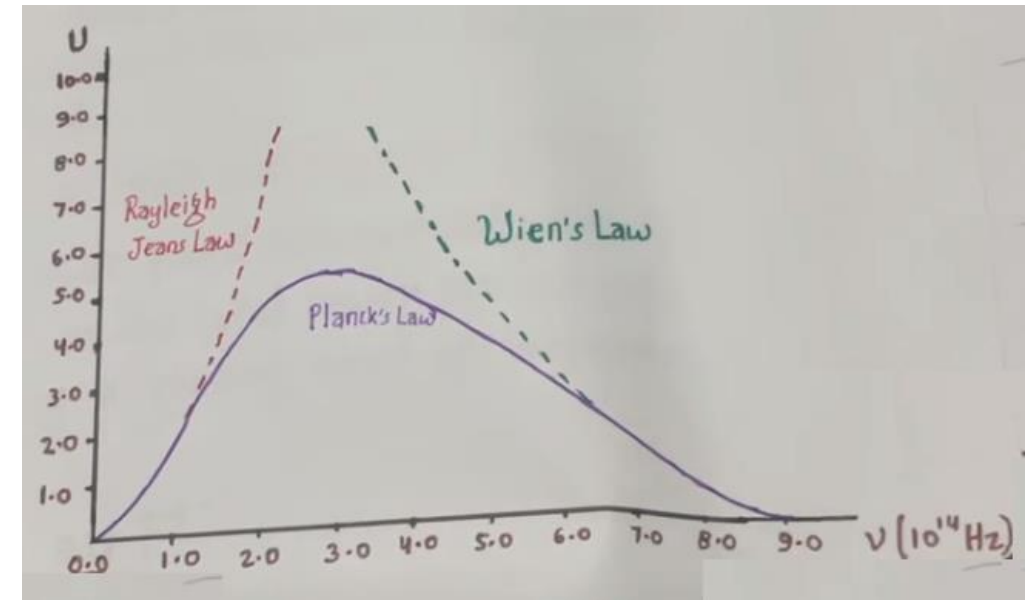
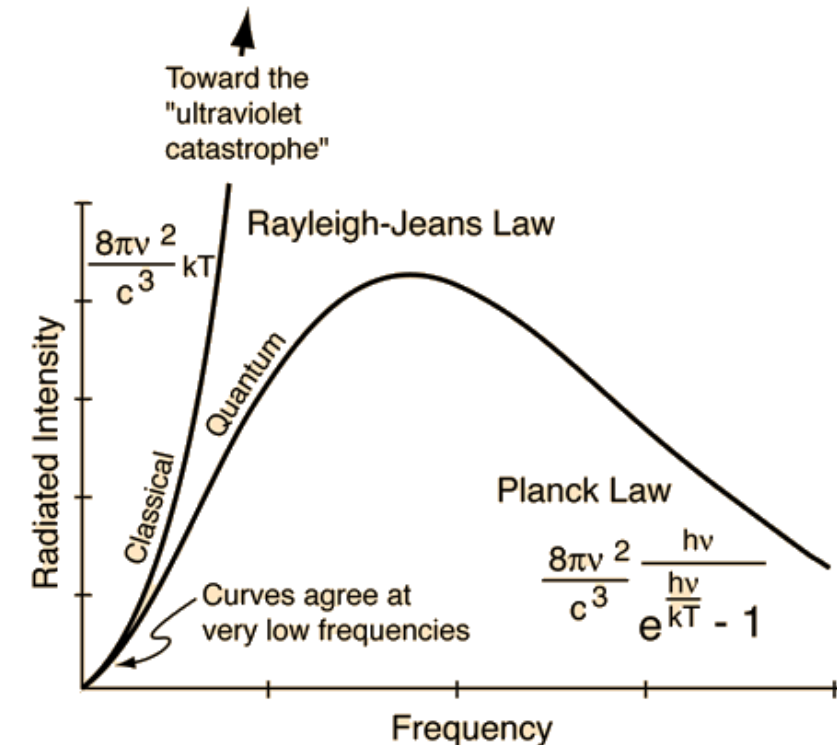


Black Body Radiation Explained By

Rayleigh-Jeans $u_{\text{RJ}}(\lambda) = \frac{8\pi k_B T}{\lambda^4}$

Wien $u_{\text{W}}(\lambda) \propto \frac{e^{-\beta/\lambda T}}{\lambda^5}$

Planck distributions $u_{\text{P}}(\lambda) = \frac{8\pi hc}{\lambda^5 (e^{hc/\lambda k_B T} - 1)}$



Planck's theory of black-body radiation

Explanation : Planck's Quantum Theory

Planck suggested that blackbody radiations can be explained if the energy of oscillating electrons (in atom) is taken as discrete rather than continuous. He derived the radiation law by using the following assumptions :

- (a) A box containing black body radiations also contains simple harmonic oscillators (or electrons) which can vibrate with all possible frequencies.
- (b) The frequency of radiation emitted by an oscillator is the same as frequency of its vibration.
- (c) An oscillator can not emit energy in a continuous manner, it can emit energy in the multiples of a small unit called quantum (photon). If an oscillator is vibrating with a frequency ν , it can only radiate in quantas of magnitude $h\nu$, i.e. the oscillator can have only discrete energy values E_n given by

$$E_n = nh\nu$$

where h = planck's constant = $6.625 \times 10^{-34} \text{ J.s}$

- (d) The oscillators (electrons) can emit or absorb radiation energy in packet of $h\nu$. This implies that the exchange of energy between radiation and matter can not take place continuously but are limited to discrete set of values $0, h\nu, 2h\nu, 3h\nu, \dots, nh\nu$.

Dual Nature of Light and Matter

❑ Light and electromagnetic radiation

- The photoelectric effect and the Compton scattering suggest particle nature of light with energy hf and momentum h/λ .
- At the same time light behaves like wave.

Light has a dual nature, exhibiting both wave and particle characteristics.

❑ Wave properties of particles

- De Broglie's idea :

All forms of matter have both properties - wave and particle characteristics

- According to de Broglie's idea, electrons like light have a dual particle-wave nature with the following relation among the momentum, energy and wavelength (de Broglie wavelength of a particle) :

$$\lambda = h / p = h / (mv)$$

An application: electron microscope

$$f = E / h$$

- His postulate was verified by diffraction pattern produced by electrons.

Only for GK

Comparison	Wave Nature of Electrons	Particle Nature of Electrons
Behavior	Interference, diffraction	Discrete, trajectories
Experiments	Diffraction, double-slit	Photoelectric effect, Compton scattering
Interference	Interference patterns	No interference
Quantization	Continuous energy	Quantized energy levels
De Broglie Wavelength	De Broglie wavelength	Less relevant
Energy Distribution	Continuous energy range	Discrete energy levels
Measurement Impact	Wavefunction collapse	No wavefunction collapse
Electron Microscopy	High-resolution imaging	High-resolution imaging
Photoelectric Effect	Not explained by waves	Demonstrates particle behavior
Compton Scattering	Explains particle behavior	Supports particle nature

Compton Effect

In 1923 A.H. Compton while studying the scattering of X-rays by a block of graphite observed that two types of X-rays were found in scattered X-rays. One, whose wavelength is same as that of incident X-rays and other, whose wavelength is greater than that of incident X-rays. This difference in wavelength, called Compton shift depends on the scattering angle. This effect is known as **Compton effect**.

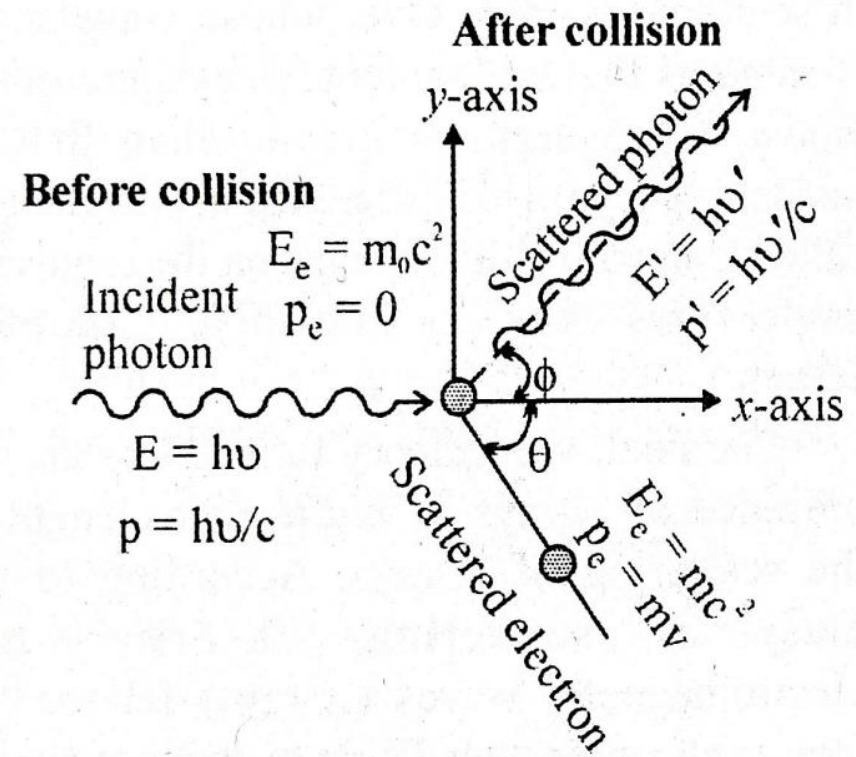


Fig. (9) : Scattering of a photon

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0c} (1 - \cos \phi)$$

Classical wave theory fails to explain the presence of X-rays of higher wavelength in the scattering of X-rays. According to the classical scattering theory when electromagnetic waves (X-rays) fall on the atomic electrons, they begin to oscillate simple harmonically and start radiating (scatter) electromagnetic waves of same frequency or wavelength as that of incident radiation. The scattering of EM waves of same frequency as that of incident radiation can be observed in all regions of electromagnetic spectrum.

Therefore according to classical scattering theory the scattered radiation must be of the same wavelength as that of incident radiation and further the scattering coefficient should be independent of wavelength. Thus classical theory cannot account for the Compton effect where the wavelength shift towards higher value in the scattered X-rays is a significant feature.

Explanation of Compton effect as evidence of quantum nature of radiation :

Compton considering the quantum nature of radiation in which radiation is constituted by energy packets of energy $h\nu$ called photons, explained the Compton effect as caused by the elastic collision between the incident photon and loosely bound electron of an atom (scatterer). The success of this theory thus provides an important evidence for the quantum nature of radiation.

In order to explain Compton effect let a photon of energy $E = h\nu$ and momentum $p = h\nu/c$ be incident on a loosely bound stationary electron whose rest energy is m_0c^2 and momentum is zero. As a result of elastic collision the electron gains momentum $p_e = mv$ and its energy becomes $E_e = mc^2$ and it recoils in a direction making an angle, say θ , with the direction of incident beam. Here m is the mass of the electron moving with a speed v .

At the same time a secondary photon of energy $E' = h\nu'$ and momentum $p' = \frac{h\nu'}{c}$ is scattered at an angle ϕ with the direction of incident beam. For all this process, it is

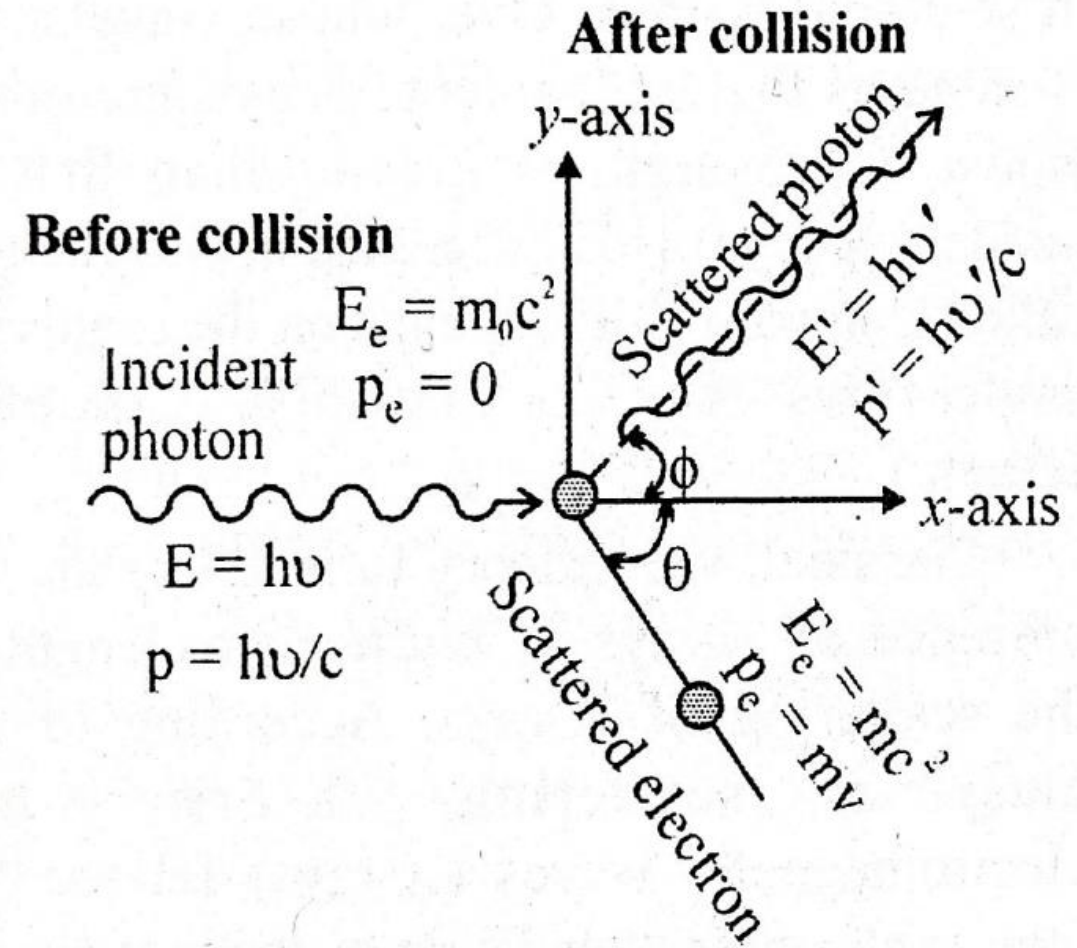


Fig. (9) : Scattering of a photon

$$(E=mc^2 \Rightarrow h\nu=mc^2, h\nu/c=mc \Rightarrow h\nu/c=P)$$

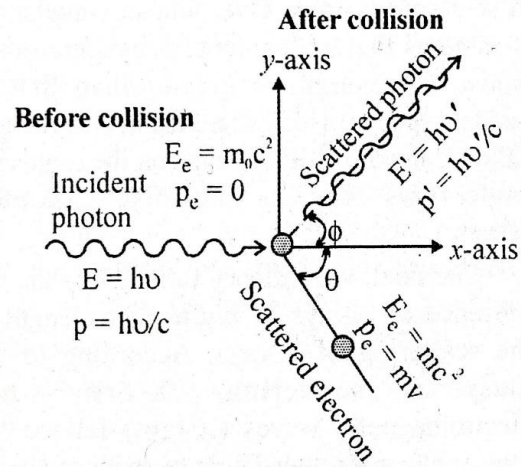


Fig. (9) : Scattering of a photon

and for the conservation of momentum in the y-direction.

$$0 = \frac{h\nu'}{c} \sin \phi - mv \sin \theta$$

$$\text{or } mvc \sin \theta = h\nu' \sin \phi \quad \dots(3)$$

Squaring equations (2) and (3) and then adding,

$$m^2v^2c^2 = h^2(\nu^2 + \nu'^2 - 2\nu\nu' \cos \phi) \quad \dots(4)$$

Squaring equation (1),

$$m^2c^4 = h^2(\nu^2 + \nu'^2 - 2\nu\nu') + m_0^2c^4 + 2hm_0c^2(\nu - \nu') \quad \dots(5)$$

Subtracting equation (4) from equation (5),

$$m^2c^4 \left(1 - \frac{v^2}{c^2} \right) = -2\nu\nu' h^2 (1 - \cos \phi) + m_0^2c^4 + 2hm_0c^2(\nu - \nu') \quad \dots(6)$$

It is assumed that principle of energy and momentum conservation hold good.

According to the principle of energy conservation,

$$h\nu + m_0c^2 = h\nu' + mc^2$$

$$\text{or } mc^2 = h(\nu - \nu') + m_0c^2 \quad \dots(1)$$

Using the principle of conservation of momentum in the x-direction,

$$\frac{h\nu}{c} = \frac{h\nu'}{c} \cos \phi + mv \cos \theta$$

$$\text{or } mvc \cos \theta = h(\nu - \nu' \cos \phi) \quad \dots(2)$$

Subtracting equation (4) from equation (5),

$$m^2 c^4 \left(1 - \frac{v^2}{c^2} \right) = -2 v v' h^2 (1 - \cos \phi) + m_0^2 c^4 + 2 h m_0 c^2 (v - v') \quad \dots\dots(6)$$

From relativistic mass formula,

$$m = \frac{m_0}{\left(1 - v^2 / c^2 \right)^{1/2}}$$

$$\text{or } m^2 \left(1 - \frac{v^2}{c^2} \right) = m_0^2$$

$\therefore$ From equation (6),

$$m_0^2 c^4 = -2 v v' h^2 (1 - \cos \phi) + m_0^2 c^4 + 2 h m_0 c^2 (v - v')$$

$$\text{or } \frac{v - v'}{v v'} = \frac{h}{m_0 c^2} (1 - \cos \phi) \quad \dots\dots(7)$$

$$\text{But } \lambda' - \lambda = \frac{c}{v'} - \frac{c}{v} = c \left(\frac{v - v'}{v v'} \right)$$

$$\therefore \Delta \lambda = \lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos \phi) \quad \dots\dots(8)$$

This relation gives Compton shift, which is independent of wavelength. The dimensions of constant $h/m_0 c$ in equation (8) is equivalent to those of wavelength, so it is called Compton wavelength λ_c .

$\therefore$ Compton wavelength

$$\begin{aligned} \lambda_c &= \frac{h}{m_0 c} \\ &= \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} \\ &= 0.0242 \text{ \AA} \end{aligned}$$

$$\therefore (\lambda' - \lambda) = 0.0242 (1 - \cos \phi) \text{ \AA} \quad \dots\dots(9)$$

(i) When $\phi = 0$, i.e. $\lambda' = \lambda$, there is no Compton shift along the direction of incident wave.

$$\therefore \Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos \phi) \dots\dots(8)$$

(ii) When $\phi = \pi/2$, $\cos \phi = 0$

and $\Delta\lambda = \lambda' - \lambda = 0.0242 \text{ \AA}$

(iii) When $\phi = \pi$, $\cos \phi = -1$

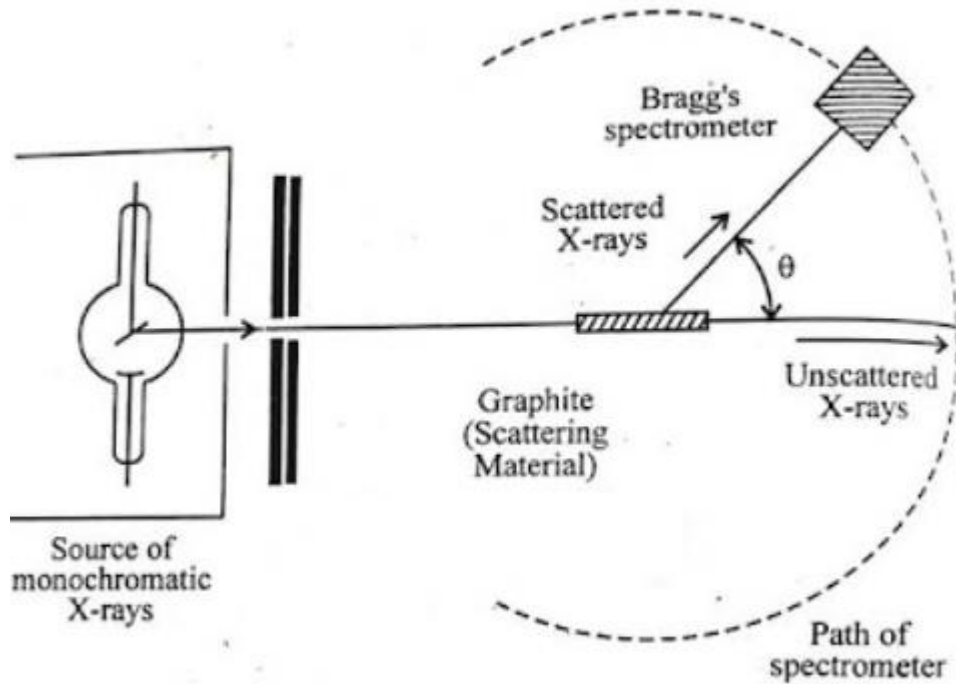
and $\Delta\lambda = \lambda' - \lambda = 0.0484 \text{ \AA}$
 $= \Delta\lambda_{\max} = 2\lambda_c$

Though the wavelength shift $\Delta\lambda$ is independent of wavelength λ but an appreciable relative change ($\Delta\lambda/\lambda$) can be observed for small values of λ . Hence using X-ray or γ -rays it is possible to study Compton effect experimentally.

The excellent agreement between the theoretical value of Compton shift and the experimental results confirms that the collision between a photon and an electron is like an

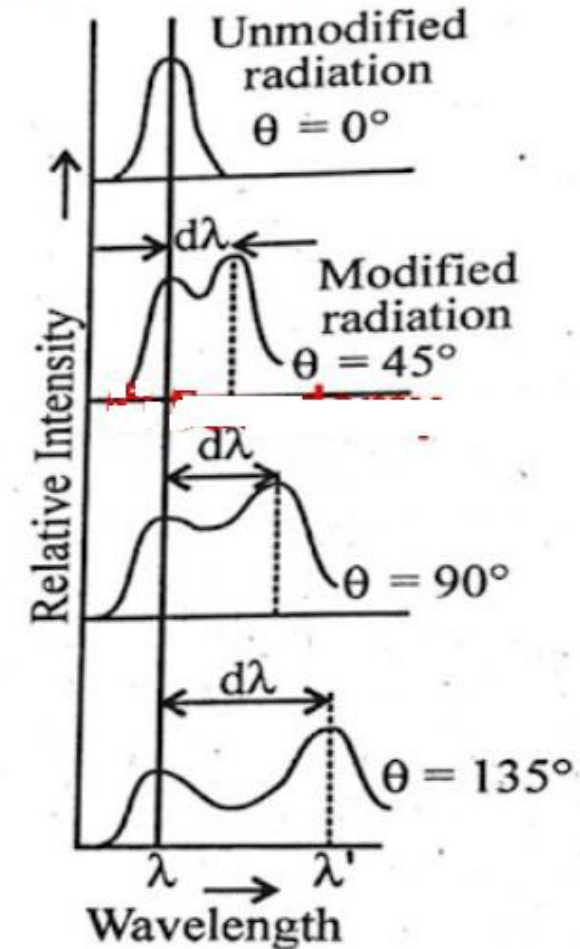
elastic collision between two particles. It proves that photon behaves like a particle. Thus Compton effect provides a direct proof of the quantum nature of radiation.

Experimental setup



Relative intensities versus wavelength curve of Compton effect at different angles

Experimental setup of Compton effect



- The excellent agreement between the Compton shift obtained from Compton theory and the experimental results confirms that the elastic collision between a photon and an electron is just like an elastic collision between two particles.
- The Compton effect proves the quantum theory (particle nature) of radiation.

Compton Effect

□ Example 27.6 : Scattering x-rays

- X-rays of wavelength $\lambda_0 = 0.200000 \text{ nm}$ are scattered from a block of material. The scattered x-rays are observed at an angle of 45.0° to the incident beam.

(a) Calculate the wavelength of the scattered x-rays at this angle.

$$\Delta\lambda = \frac{h}{m_e c} (1 - \cos\theta) = \frac{6.63 \times 10^{-34} \text{ J}\cdot\text{s}}{(9.11 \times 10^{-31} \text{ kg})(3.00 \times 10^8 \text{ m/s})} (1 - \cos 45.0^\circ) = 7.11 \times 10^{-13} \text{ m}$$

$$\lambda = \Delta\lambda + \lambda_0 = 0.200711 \text{ nm}$$

(b) Compute the fractional change in the energy of a photon in the collision.

$$E = hf = h \frac{c}{\lambda}$$

$$\frac{\Delta E}{E} = \frac{E_f - E_i}{E_i} = \frac{hc / \lambda_f - hc / \lambda_i}{hc / \lambda_i} = -\frac{\Delta\lambda}{\lambda_i} = -\frac{0.000711 \text{ nm}}{0.200711 \text{ nm}} = -3.54 \times 10^{-3}$$

Recapitulation

- Quantum theory had its origin in the attempts made to explain these results. Radically new concepts were required to explain them like *the particle nature of radiation, the wave nature of matter and the quantization of physical quantities.*
- Max Planck (1900) introduced the concept of quantization of radiant energy to explain the spectral distribution of radiant energy emitted by a heated blackbody. (Quantization of Energy) (Nobel prize in 1918)
- Niels Bohr (1913) used Planck's quantum hypothesis and gave the first successful theory to explain the stability and observed spectrum of the hydrogen atom. (Quantization of angular momentum) (Nobel prize in 1922)

Recapitulation

- Albert Einstein (1905) used Planck's idea to explain the puzzling features of the photoelectric effect. **(Quantization of EM Waves)** (**Nobel prize in 1921**)
- Franck & Hertz (1914) demonstrated the controlled excitation of atoms and molecules by hitting it with electrons of variable energy. **(Evidence of discrete energy states of atoms)**
- A.H. Compton (1923) explained the change of wavelength of X-rays when they are scattered by free or weakly bound electrons (Compton Effect).
- The works of Einstein (1905) and Compton (1923) showed that radiation can behave as particles, thus bringing in wave-particle duality of radiation.

Recapitulation

- Louis de Broglie (1924) suggested that like radiation, matter also has dual nature, i.e.. there is a wave associated with every material particle.
(Wave nature of Particle)
- The wave nature of electrons was verified by Davison and Germer and independently by G.P. Thomson in 1927.
- Schrödinger (1926) discovered the equation for matter waves and thus formulated wave mechanics.
- The wave particle duality of matter led to a very important principle, called the uncertainty principle discovered by Heisenberg in 1927.

Electrons are Particles or Waves ?

- **JJ Thomson won the Nobel Prize for Physics in 1906 for demonstrating that the electron is a particle.**
- **GP Thomson (son of JJ Thomson) won it in 1937 for demonstrating that the electron is a wave.**

Both were right

Electron has wave-particle nature

Wave- Particle Duality was family business of Thomson's family.

DE-BROGLIE MATTER WAVES

The wave theory of light successfully explained the phenomena such as **interference, diffraction and polarization** but it is failed to explain the photoelectric effect, compton effect, **absorption and emission of radiation by substances**. However these phenomena have been explained by **quantum theory of light**. According to quantum theory, light of frequency ν is composed of tiny discrete packets or bundles of energy called "**photons**", each having an **energy $h\nu$** , momentum $\frac{h\nu}{c}$ and velocity c . Photons behave almost exactly like a material particle. Thus light has a dual nature, i.e. it possess both **particle and wave properties***.

In analogy with dual character of light, **Louis de Broglie in 1924** introduced the hypothesis that **all material particles in motion possess a wave character** means matter also possesses dual character. Such waves associated with the material particles are known as "**matter waves**" or "**de Broglie waves**". All the fundamental particles as electrons, protons, neutrons, atoms and molecules etc. have waves associated with them.

De Broglie hypothesis :-

➤ Louis de Broglie (1924) suggested that like radiation, matter also has dual nature, i.e. there is a wave associated with every material particle. **(Wave nature of Particle)**
(Nobel prize in 1929)

de Broglie relation

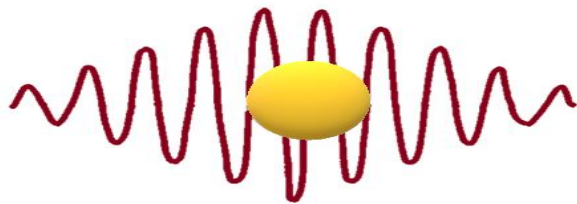
de Broglie wavelength

$$\lambda = \frac{h}{p}$$

Planck's constant

$$h = 6.63 \times 10^{-34} \text{ Js}$$

Where 'h' is planck's constant, 'p=mv' is momentum



1892-1987

“According to the de-Broglie hypothesis, “A moving particle, whatever its nature be, has wave property associated with it” and a moving material particle behaves like a wave at times and like a particle at other times. The waves associated with these particles are known as matter waves or de Broglie waves.

de Broglie Matter Waves

(i) de Broglie wavelength of a material particle of kinetic energy K

$$\therefore \text{Kinetic energy} = K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$\therefore P = \sqrt{2mK}$$

$\therefore$ Associated de Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mK}}$$

de Broglie Matter Waves

(ii) de Broglie wavelength associated with a charged particle (q), accelerated through a potential difference V volts (Non-relativistic case).

If a charged particle q accelerated from rest through a potential difference V, then kinetic energy of it is

$$K.E. = K = \frac{1}{2}mv^2 = qV$$

Therefore, the de Broglie wavelength associated with it is given by

$$\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mK}}$$
$$\lambda = \frac{h}{\sqrt{2mqV}}$$

de Broglie Matter Waves

(a) de Broglie wavelength associated with electron (Non relativistic case):

If an electron of rest mass m_e and charge e is accelerated by a potential V volts from rest to velocity v then the de Broglie wavelength associated with it is

$$\lambda = \frac{h}{\sqrt{2m_e eV}}$$

$$h = 6.626 \times 10^{-34} \text{Joule} - \text{sec}$$

$$m_e = 9.1 \times 10^{-31} \text{kg (For Non - relativistic case)}$$

$$e = 1.6 \times 10^{-19} \text{C}$$

Substituting them

$$\lambda = \frac{12.27}{\sqrt{V}} \text{\AA}$$

de Broglie Matter Waves

**(b) de Broglie wavelength associated with proton accelerated through V volts
(Non relativistic case)**

de Broglie wavelength associated with proton accelerated through V volts is

$$\lambda = \frac{h}{\sqrt{2m_p eV}}$$

$$m_p = 1.67 \times 10^{-27} \text{ kg}, \quad e = 1.6 \times 10^{-19} \text{ C}$$

Substituting the values

$$\lambda = \frac{0.286}{\sqrt{V}} \text{ \AA}$$

de Broglie Matter Waves

(iii) de Broglie wavelength associated with relativistic particles

If the velocity of a particle is comparable to that of light, then relativistic mass is given by

$$m = \frac{m_o}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where m_o is the rest mass of the particle then corresponding de Broglie wavelength is

$$\lambda = \frac{h}{mv} = \frac{h \left(\sqrt{1 - \frac{v^2}{c^2}} \right)}{m_o v}$$

If E is the total energy of a relativistic particle then from Energy-momentum relation

$$E^2 = P^2 c^2 + m_o^2 c^4$$

$$\Rightarrow Pc = \sqrt{E^2 - m_o^2 c^4}$$

de Broglie Matter Waves

Hence corresponding wavelength is

$$\lambda = \frac{h}{P} = \frac{hc}{Pc} = \frac{hc}{\sqrt{E^2 - m_o^2 c^4}}$$

But the total energy of a relativistic particle is the sum of the kinetic energy of the particle (K) and the rest mass energy $m_o c^2$, i.e.,

$$E = K + m_o c^2$$

- Putting the value of E, we get

$$\lambda = \frac{hc}{\sqrt{(K + m_o c^2)^2 - m_o^2 c^4}}$$
$$\lambda = \frac{hc}{\sqrt{K^2 + 2K m_o c^2}} \Rightarrow \lambda = \frac{hc}{\sqrt{K(K + 2m_o c^2)}}$$

de Broglie Matter Waves

(iv) de Broglie wavelength material particle in thermal equilibrium at temperature 'T'

The K.E. of material particles in thermal equilibrium at temperature T is

$$K = \frac{3}{2}kT$$

Where $k = 1.38 \times 10^{-23} \text{ Joule/K}$ is the Boltzmann constant

Associated de Broglie wavelength

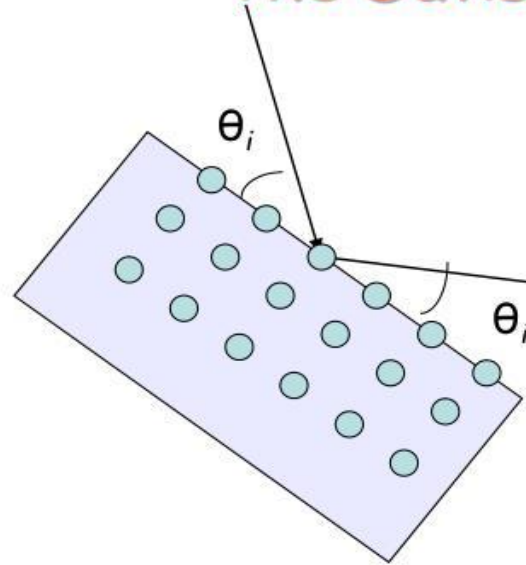
$$\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mK}}$$
$$\lambda = \frac{h}{\sqrt{2m \times \frac{3}{2}KT}} = \frac{h}{\sqrt{3mKT}}$$

Concept of Quantization

Development of Quantum Mechanics

ELECTRON DIFFRACTION

The Davisson-Germer experiment (1927)



The Davisson-Germer experiment: scattering a beam of electrons from a Ni crystal. Davisson got the 1937 Nobel prize.

Davisson

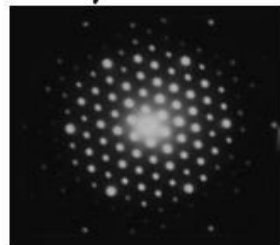


G.P.

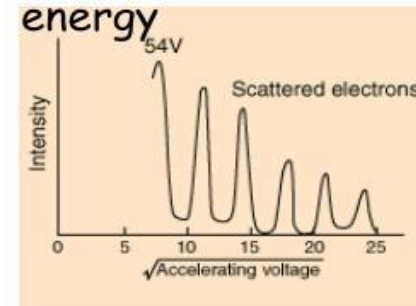
Thomson



At fixed accelerating voltage (fixed electron energy) find a pattern of sharp reflected beams from the crystal



At fixed angle, find sharp peaks in intensity as a function of electron energy



Davisson, C. J., "Are Electrons Waves?," Franklin Institute Journal 205, 597 (1928)

G.P. Thomson performed similar interference experiments with thin-film samples

➤ The wave nature of electrons was verified by Davison and Germer and independently by G.P. Thomson in 1927.

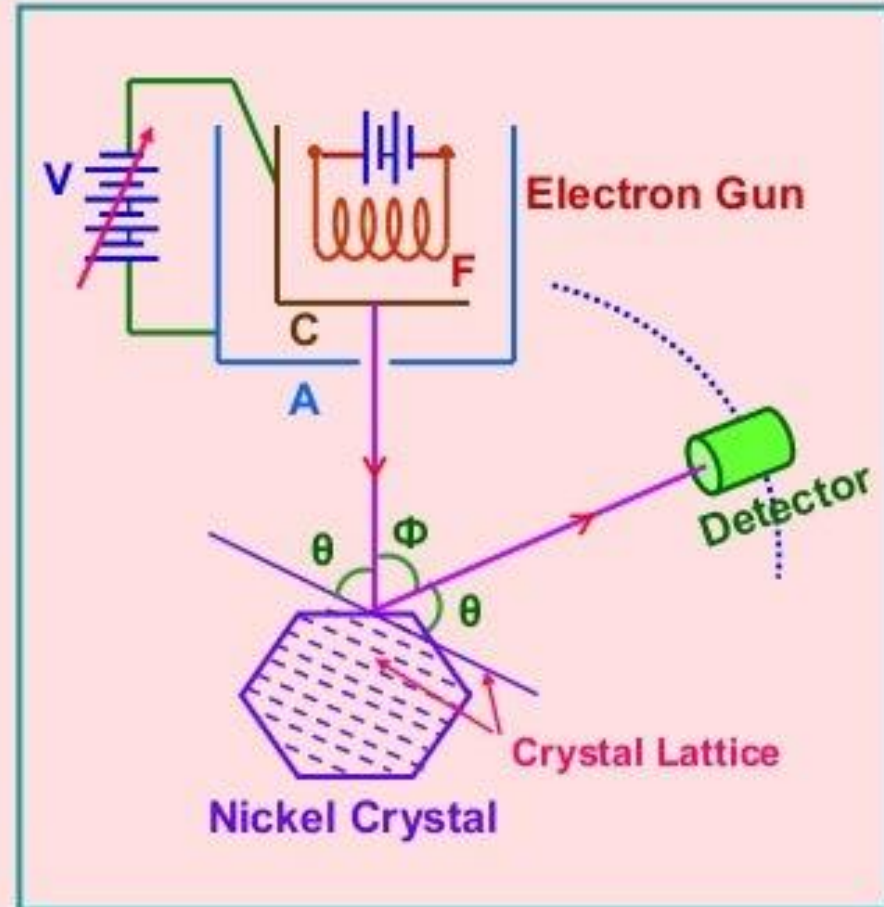
Davisson and Germer Experiment:

A beam of electrons emitted by the electron gun is made to fall on Nickel crystal cut along cubical axis at a particular angle.

The scattered beam of electrons is received by the detector which can be rotated at any angle.

The energy of the incident beam of electrons can be varied by changing the applied voltage to the electron gun.

Intensity of scattered beam of electrons is found to be maximum when angle of scattering is 50° and the accelerating potential is 54 V .



$$\theta + 50^\circ + \theta = 180^\circ \quad \text{i.e.} \quad \theta = 65^\circ$$

For Ni crystal, lattice spacing
 $d = 0.91\text{ \AA}$

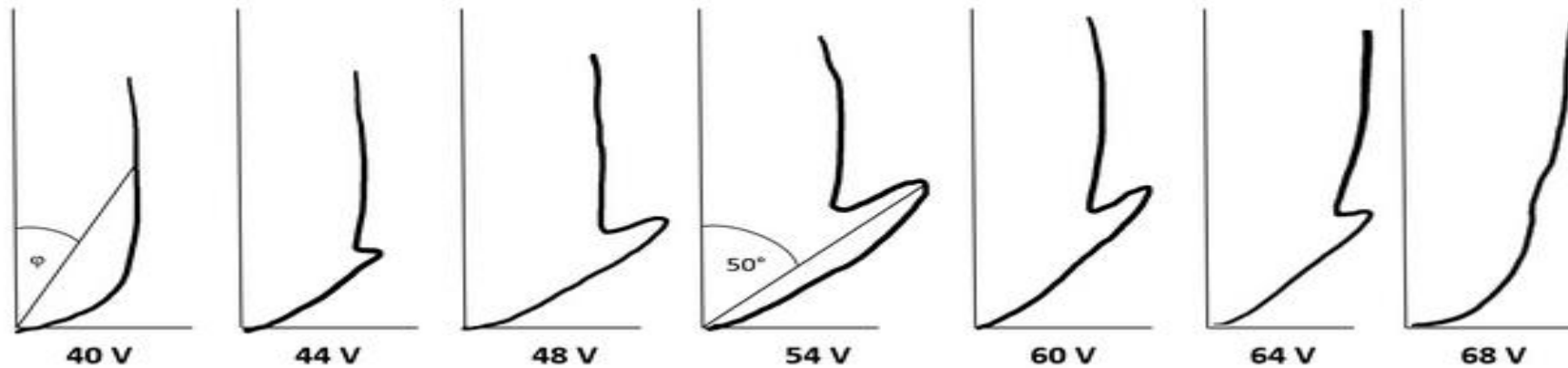
For first principal maximum, $n = 1$

Electron diffraction is similar to X-ray diffraction.

∴ Bragg's equation $2d\sin\theta = n\lambda$ gives

$$\lambda = 1.65\text{ \AA}$$

The experimental curves obtained by Davisson and Germer are as shown in figure.

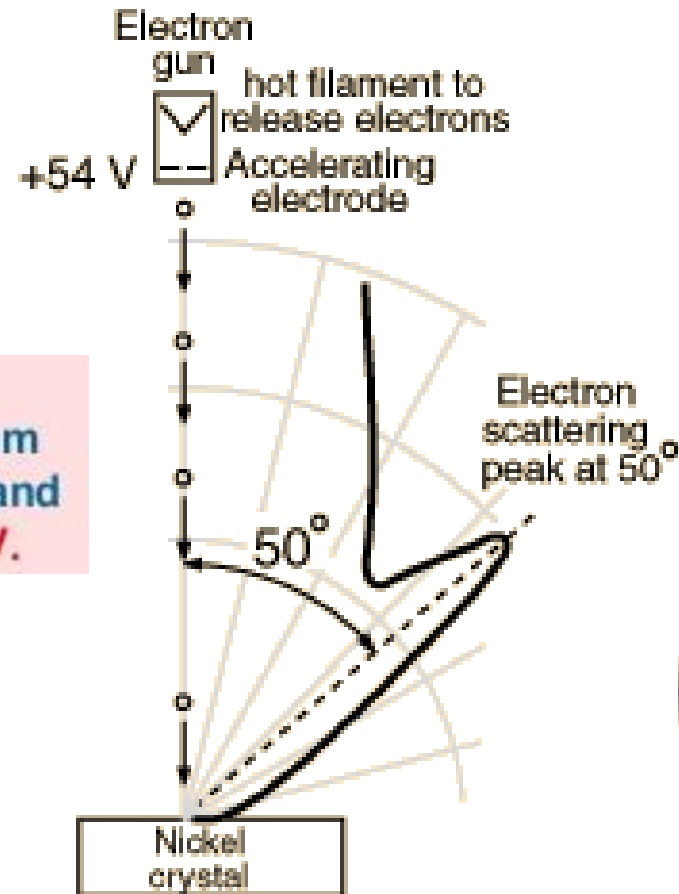


From these experimental curves, the following inferences can be drawn :

- (i) The intensity of scattered electrons depends upon the angle of scattering ϕ .
- (ii) Always a 'bump' or a kink occurs in the curve at $\phi = 50^\circ$, the angle which the scattered beam makes with the incident beam.
- (iii) The size of the bump goes on increasing as the accelerating voltage is increased.
- (iv) The size of the bump becomes maximum, when the accelerating voltage is 54 volt.
- (v) The size of the bump starts decreasing with further increase in the accelerating voltage.

However, according to classical physics, there should be very little variation in the intensity of electron beam with the angle of scattering voltages. The appearance of bump in a particular direction is due to constructive interference of electrons scattered from different layers of regularly spaced atoms of the crystal. This establishes the wave nature of light.

Intensity of scattered beam of electrons is found to be maximum when angle of scattering is **50°** and the accelerating potential is **54 V**.



Theory

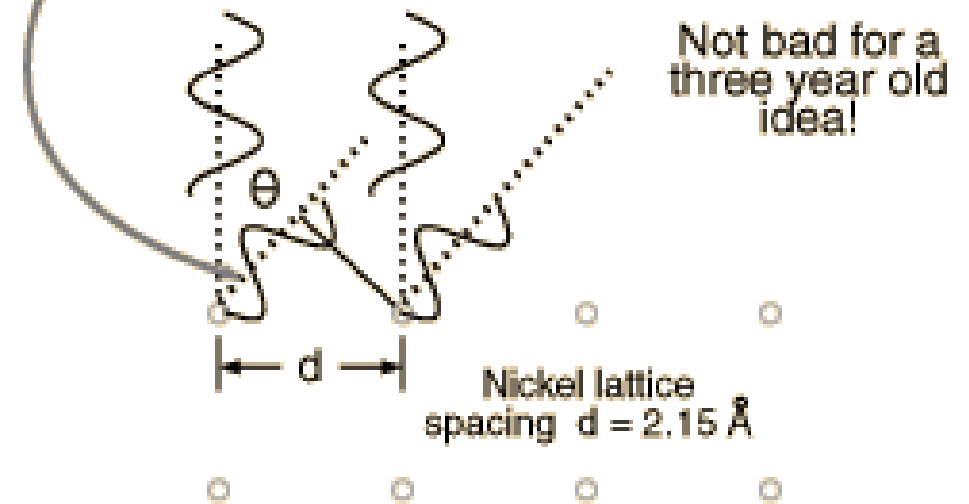
$$\lambda = \frac{h}{mv} = 1.67 \text{ \AA} \text{ for } 54 \text{ V}$$

Experiment

Pathlength difference

$$d \sin \theta = 2.15 \sin 50^\circ = \lambda = 1.65 \text{ \AA}$$

for constructive interference



1924
de Broglie's
hypothesis

1927
Davisson-
Germer
experiment

1929
Nobel Prize
for
de Broglie

These experiments by Davisson and Germer and by Thomson proved that de Broglie's waves are not simply mathematical conveniences, but have observable physical effects. The 1937 Nobel Prize in Physics was awarded to these gentlemen for their pioneering work. Just as Compton showed that waves could act like particles, Davisson and Germer showed that particles could act as waves.

Wave Nature Of Matter $\rightarrow$ Matter Waves

electron. (small particles)

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$v = 10^6 \text{ m/s}$$

$$\lambda \approx 7.33 \text{ \AA}$$

Significant

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{m v}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^6}$$

$$\approx \frac{6.6}{9.1} \times 10^{-34+31-6}$$

$$\approx 7.33 \times 10^{-10} \text{ m}$$

$$\approx 7.33 \text{ \AA}$$

For partial m-100KG, V=10m/s

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{100 \times 10}$$

$$\lambda = 6.626 \times 10^{-37} \text{ m}$$

$$\lambda = 6.626 \times 10^{-27} \text{ \AA}$$

Very very small λ
Cannot be detected

UNCERTAINTY PRINCIPLE

Some-times moving material particle like electron, proton, neutron etc. behaves like a wave as well as a particle. So while containing wave-Particle dual character of matter, is it possible to determine the exact position of a particle in space at some given instant ? It is answered by **Heisenberg** in 1927 stating "Uncertainty principle" as follows :

Statement:- It is not possible to make a simultaneous determination of the position and the momentum of a particle in a given direction with unlimited precision. Mathematical representation of Heisenberg uncertainty principle is

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} * \quad \text{....(4.40)}$$

Generally $\Delta x \Delta p_x \sim \hbar$

$$\text{where } \hbar = \frac{h}{2\pi} = 1.05 \times 10^{-34} \text{ J-s}$$

where Δx is the uncertainty in the position in x -direction and Δp_x is the uncertainty in the momentum in x -direction

If an experiment is designed for the exact determination of one of them, the determination of the other will become completely uncertain.

If a particle momentum is known accurately $\Rightarrow \Delta p_x = 0$ then using equation (4.40) $\Delta x \approx \infty$, i.e., uncertainty in the position is infinite which means we can't determine the position and that particle is represented by a wave of constant amplitude extending over all space.

Hence it is impossible to know both the exact position and exact momentum in a given direction of a particle at the same time.

Heisenberg uncertainty principle can also be stated as it is not possible to make a simultaneous determination of the energy and the time coordinate of a particle with unlimited precision. Mathematically it can be represented as

$$\Delta E \cdot \Delta t \geq \hbar \quad \dots(4.41)$$

where ΔE is the uncertainty in energy of a particle and Δt is the uncertainty in time.

Application of Uncertainty Principle

The principle of uncertainty explains a large number of facts which could not be explained by classical ideas.

The non-existence of the electron in the nucleus and the existence of the protons or neutron in nucleus.

The radius of the nucleus of any atom is of the order of $10^{-14}m$, so that if a particle confined with nucleus, the uncertainty in its position (Δx) must not be greater than $10^{-14}m$, i.e., $\Delta x \approx 2 \times 10^{-14}m$.

Using the uncertainty principle, the uncertainty in the momentum of particle is given by

$$\Delta p_x \approx \frac{\hbar}{\Delta x}$$

$$\Delta p_x \approx \frac{1.055 \times 10^{-34} \text{ J-s}}{2 \times 10^{-14} \text{ m}}$$

$$\Delta p_x \approx 5.275 \times 10^{-21} \text{ Kg.m/s}$$

If this is the uncertainty in momentum of the particle then the momentum component p_x and hence the total momentum p of particle in the nucleus must be at least of the order of magnitude $|p|$, i.e.,

$$|\vec{p}| \approx p_x \approx 5.275 \times 10^{-21} \text{ Kg.m/s}$$

(i) For electron –

Since the magnitude of momentum is relativistic compared to the mass of electron $m_0 = 9.1 \times 10^{-31}$ kg., then using the relativistic formula for the energy of the electron

$$E^2 = p^2 c^2 + m_0^2 c^4$$

As the rest energy $m_0 c^2$ of the electron is of the order of 0.511 MeV, which is much smaller than the value of first term, hence neglecting this term we get

$$E^2 = p^2 c^2$$

$$E = pc = (5.275 \times 10^{-21}) \times (3 \times 10^8) J$$

$$E = \frac{5.275 \times 10^{-21} \times 3 \times 10^8}{1.6 \times 10^{-19}} \text{ eV}$$

$$E \approx 10 \text{ MeV}$$

Thus, the electron must have an energy of the order of 10 MeV if it is to be a nuclear constituent. But since the energy of a β -particle (electron) emitted from the nucleus has not been found to exceed 2–3 MeV, we conclude that electrons does not exist within nuclei.

(ii) For proton or neutron –

This will be a non-relativistic case due to mass of proton or neutron

$$m_0 \approx 1.67 \times 10^{-27} \text{ kg} \quad \therefore \quad |\vec{v}| = \frac{|\vec{p}|}{m_0} \approx 3 \times 10^6 \text{ m/s}$$

The kinetic energy of proton or neutron

$$= K = \frac{p^2}{2m_0}$$

$$K = \frac{(5.275 \times 10^{-21} \text{ kg-m/s})^2}{2 \times 1.67 \times 10^{-27} \text{ kg}}$$
$$= \frac{(5.275 \times 10^{-21})^2}{2 \times 1.67 \times 10^{-27} \times 1.6 \times 10^{-19}} \text{ eV}$$

$$K \approx 52 \text{ KeV}$$

Since this energy is smaller than the energies carried by these particles (protons or neutrons) emitted by nuclei, hence we conclude that both these particles can exist inside the nuclei.

Postulates of Wave Mechanics

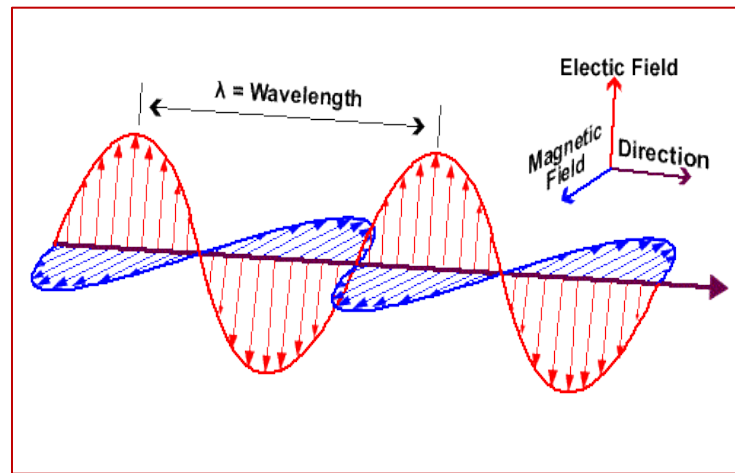
In the development of quantum mechanics, there are certain basic postulates and essential definitions which are of fundamental importance.

Postulate I: The quantum state of a particle is characterized by a wave function, $\psi(\vec{r}, t)$ a complex function of space and time, which contains all the information it is possible to obtain about the particle.

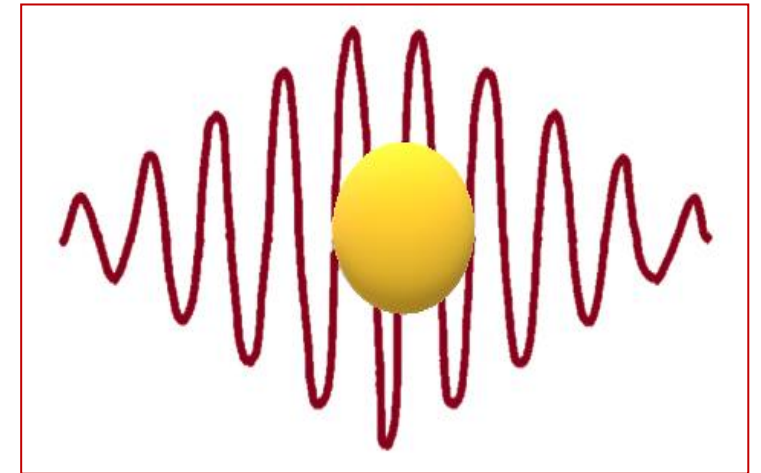
The wave function is interpreted as a probability amplitude of particle's presence.



Sound waves



Electromagnetic waves



Matter waves

Postulates of Wave Mechanics

Postulate II: The wave function $\psi(\vec{r}, t)$ is the probability amplitude for the particle's presence. That is the probability of observing the particle at time t in the volume element dV is

$$P(\vec{r}, t) = |\psi(\vec{r}, t)|^2 dV = \psi^*(\vec{r}, t) \psi(\vec{r}, t) dV$$

Diagram illustrating the components of the probability equation:

- $P(\vec{r}, t)$ is labeled **Probability** (indicated by a green arrow pointing to the left).
- $|\psi(\vec{r}, t)|^2$ is labeled **Wave function** (indicated by a green arrow pointing down).
- $\psi^*(\vec{r}, t)$ is labeled **Complex conjugate of $\psi(\vec{r}, t)$** (indicated by a green arrow pointing down).
- dV is labeled **Volume element** (indicated by a green arrow pointing down).

Because of uncertainty principle we cannot define the position of particle so we have to work with probabilities.

Since the probability of finding a particle in whole universe is one i.e.

$$\int_{-\infty}^{+\infty} |\psi(\vec{r}, t)|^2 dV = \int_{-\infty}^{+\infty} \psi^*(\vec{r}, t) \psi(\vec{r}, t) dV = 1$$

This is Normalization condition and the wave function is said to be normalized.

Postulates of Wave Mechanics

Postulate III: Every observable quantity can be expressed by an operator in quantum mechanics. An operator for the observable quantity is anything, any mathematical operation as multiplication, differentiation or integration, which is capable of operating on a wave function and extract the value of observable quantity.

$$\hat{A}\psi(\vec{r},t) = a\psi(\vec{r},t)$$

Diagram illustrating the eigenvalue equation $\hat{A}\psi(\vec{r},t) = a\psi(\vec{r},t)$ with labels and arrows:

- Operator**: Points to $\hat{A}$
- Eigen function**: Points to $\psi(\vec{r},t)$ (under the hat)
- Eigen value**: Points to a
- Eigen function**: Points to $\psi(\vec{r},t)$ (after the equals sign)

Physical Quantity $\rightarrow$ Operators

symbol $\leftrightarrow$ actual operation

Momentum $p_x \rightarrow \hat{p}_x \leftrightarrow \frac{\hbar}{i} \frac{\partial}{\partial x}$

Total Energy $E \rightarrow \hat{E} \leftrightarrow i\hbar \frac{\partial}{\partial t}$

Coordinate $x \rightarrow \hat{x} \leftrightarrow x$

Potential Energy $U(x) \rightarrow \hat{U}(x) \leftrightarrow U(x)$

Postulates of Wave Mechanics

Postulate IV: When a measurement of an observable quantity 'f' is made on a system or for which the wave function is $\psi(\vec{r}, t)$, we get different values of f during different trials. This is in conformity with the uncertainty principle. The most probable value of f is given by

$$\langle f \rangle = \frac{\int_{-\infty}^{\infty} \psi^*(\vec{r}, t) \hat{F} \psi(\vec{r}, t) dV}{\int_{-\infty}^{\infty} \psi^*(\vec{r}, t) \psi(\vec{r}, t) dV}$$

Where $\hat{F}$ is the operator associated with observable quantity 'f' and $\psi^*(\vec{r}, t)$ is complex conjugate of wave function $\psi(\vec{r}, t)$. The quantity $\langle f \rangle$ is called the expectation value or average value of the observable quantity 'f'.

If the wave function is normalized, then hence the expectation value of 'f'

$$\langle f \rangle = \int_{-\infty}^{\infty} \psi^*(\vec{r}, t) \hat{F} \psi(\vec{r}, t) dV$$

Postulates of Wave Mechanics

Postulate V: A particle moving in space can be represented by the following wave equation

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}, t) + U(\vec{r}, t) \psi(\vec{r}, t)$$

$\psi(\vec{r}, t)$ = wave function

m= mass of the particle

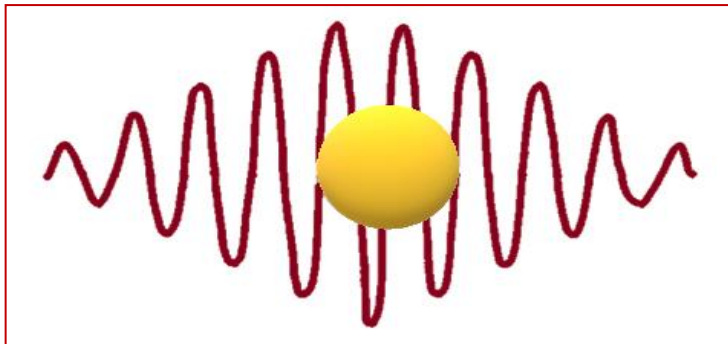
$U(\vec{r}, t)$ = potential energy of particle at position $\vec{r}$ at any instant t

This is called the Schrödinger wave equation. This equation is applicable to all cases of particle motion and the nature of potential decides what solutions hold for the given case.



Wave Function :-

- Wave function $\psi(\vec{r}, t)$ is the quantity whose variations make up the matter waves.
- It is a complex function of space and time, which contains all the information it is possible to obtain about the particle.
- The amplitude of matter wave is described by wave function $\psi(\vec{r}, t)$.
- The value of wave function $\psi(\vec{r}, t)$ associated with a particular point $\vec{r}$ in space at the time 't' is related to the probability of finding the particle there at that time.



$$\psi(\vec{r}, t) = Ae^{-i(\omega t - \vec{k} \cdot \vec{r})}$$

Physical Significance of Wave Function $\psi(\vec{r}, t)$

- The wave function $\psi(\vec{r}, t)$, which is a solution of Schrödinger equation, provides a complete description of the behaviour of a particle in a potential field $U(\vec{r}, t)$.
- $\psi(\vec{r}, t)$ is a complex function, which itself has no physical meaning but it can be made related to the probability of finding the particle at the position $\vec{r}$ at the time t .
- Max Born in 1926 proposed a meaning to the wave function $\psi(\vec{r}, t)$. According to him $\psi(\vec{r}, t)$ may be regarded as a measure of the probability of finding the particle at a particular position $\vec{r}$ and a particular time t .
- The wave function $\psi(\vec{r}, t)$ is usually a complex quantity but the probability has to be real and positive.

Physical Significance of Wave Function $\psi(\vec{r}, t)$

- The probability density for a complex wave function $\psi(\vec{r}, t)$ is therefore taken as the square of the absolute magnitude of $\psi(\vec{r}, t)$
Thus the probability density

$$= |\psi(\vec{r}, t)|^2 = \psi^*(\vec{r}, t)\psi(\vec{r}, t)$$

And the probability of finding the particle in the volume dV .

$$P(\vec{r}, t) = |\psi(\vec{r}, t)|^2 dV = \psi^*(\vec{r}, t)\psi(\vec{r}, t)dV$$

Physical Significance of Wave Function $\psi(\vec{r}, t)$

- **Normalization of $\psi(\vec{r}, t)$** : As it is certain that for a given particle somewhere in space the total probability of finding the particle somewhere must be unity (100%). Hence

$$\int_{-\infty}^{+\infty} |\psi(\vec{r}, t)|^2 dV = \int_{-\infty}^{+\infty} \psi^*(\vec{r}, t) \psi(\vec{r}, t) dV = 1$$

This condition is known as Normalization condition and the wave function $\psi(\vec{r}, t)$ is said to be *normalized*.

Condition of Orthogonality

- **Orthogonality of $\psi(\vec{r}, t)$** : Let $\psi_m(\vec{r}, t)$ and $\psi_n(\vec{r}, t)$ are two different wave functions satisfying the wave equation for a given system. If $\psi_n(\vec{r}, t)$ and $\psi_m(\vec{r}, t)$ are such that the integral $\int \psi_m^*(\vec{r}, t) \psi_n(\vec{r}, t) dV$ and $\int \psi_n^*(\vec{r}, t) \psi_m(\vec{r}, t) dV$ vanishes over entire space, i.e.

$$\int_{-\infty}^{+\infty} \psi_m^*(\vec{r}, t) \psi_n(\vec{r}, t) dV = \int_{-\infty}^{+\infty} \psi_n^*(\vec{r}, t) \psi_m(\vec{r}, t) dV = 0$$

Then these wave functions are said to be mutually orthogonal. This condition is known as **Condition of Orthogonality**.

Properties of Wave Function $\psi(\vec{r}, t)$

$\psi(\vec{r}, t)$ must be well behaved in the given region i.e.,

➤ **Wave function along with its derivatives must be single valued.**

If $\psi(\vec{r}, t)$ has more than one value at any point, it means more than one value of probability of finding the particle at that point which is obviously ridiculous.

➤ **Wave function along with its derivatives must be finite everywhere.**

If $\psi(\vec{r}, t)$ is finite for a particular point, it means an infinite large probability of finding the particles at that point. This would violate the uncertainty principle.

Properties of Wave Function $\psi(\vec{r}, t)$

- Wave function along with its derivatives must be continuous everywhere.

As the derivative of a discontinuous function is always infinite which does not fulfill second condition.

- Wave function $\psi(\vec{r}, t)$ must be normalized i.e.,

$$\int_{-\infty}^{+\infty} |\psi(\vec{r}, t)|^2 dV = \int_{-\infty}^{+\infty} \psi^*(\vec{r}, t) \psi(\vec{r}, t) dV = 1$$

- $\psi(\vec{r}, t)$ must obey superposition principle. If ψ_1 & ψ_2 are two solutions of Schrödinger's equation then $\psi_1 + \psi_2$ is also a solution.

Schrödinger's equation

The idea of matter waves was used by Schrödinger to arrive at a wave equation that describes quantum mechanical particles like electron, proton etc. as a wave that is propagating disturbance i.e., disturbance that varies periodically with position $\vec{r}$ and time t .

Just as varying electric field E contains all the information about electromagnetic waves, Schrödinger defined a wave function which contains all the information about the behavior of the particle.

$$i\hbar \frac{\partial \psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}, t) + U(\vec{r}, t) \psi(\vec{r}, t)$$



1887-1961

Schrödinger (1926) discovered the equation for matter waves and thus formulated wave mechanics.

Time-dependent Schrödinger's equation

Since matter waves are not real waves, the wave function is expected to be complex in general and therefore the acceptable wave function for matter waves may be written by a simple guess

$$\psi(x, t) = Ae^{-i(\omega t - kx)} \quad \text{.....(1)}$$

Where one-dimensional case has been taken for convenience and may be generalized to three dimension from symmetry considerations. Angular frequency ω for matter waves is now defined as

$$\omega = 2\pi\nu = \frac{2\pi h \nu}{h} = \frac{E}{\hbar} \quad \text{.....(2)}$$

Schrödinger's equation

Time dependent case

and the wave vector k is defined as

$$k = \frac{2\pi}{\lambda} = \frac{2\pi\hbar}{h\lambda} = \frac{p}{\hbar} \quad \text{.....(3)}$$

where E is the total energy of the particle and p is its momentum.

If particle moves with speed that is small as compared to that of light, the total energy of the particle E is sum of kinetic energy $p^2/2m$ and potential energy $U(x,t)$. Where $U(x,t)$ is in general function of x and t .

Thus,

$$E = \frac{p^2}{2m} + U(x,t) \quad \text{.....(4)}$$

Potential energy $U(x,t)$ represents the influence of rest of the universe on the particle.

Schrödinger's equation

(Time-dependent case)

$$\psi(x, t) = Ae^{-i(\omega t - kx)} \quad \dots(1)$$

$$E = \frac{p^2}{2m} + U(x, t) \quad \dots(4)$$

Multiplying Eq. (4) on both sides by $\psi(x, t)$, we obtain

$$E \psi(x, t) = \frac{p^2}{2m} \psi(x, t) + U(x, t) \psi(x, t) \quad \dots(5)$$

Differentiating eq. (1) twice with respect to x

$$\frac{\partial \psi(x, t)}{\partial x} = (ik) Ae^{-i(\omega t - kx)}$$

$$\frac{\partial^2 \psi(x, t)}{\partial x^2} = (ik)^2 \psi(x, t)$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi\hbar}{h\lambda} = \frac{p}{\hbar} \quad \dots(3)$$

$$\Rightarrow \frac{\partial^2 \psi(x, t)}{\partial x^2} = -k^2 \psi(x, t) \Rightarrow \frac{\partial^2 \psi(x, t)}{\partial x^2} = -\frac{p^2}{\hbar^2} \psi(x, t) \quad \text{Since } k = \frac{p}{\hbar}$$

$$\Rightarrow p^2 \psi(x, t) = -\hbar^2 \frac{\partial^2 \psi(x, t)}{\partial x^2} \quad \dots(6)$$

Schrödinger's equation

Time dependent case

$$\psi(x, t) = Ae^{-i(\omega t - kx)} \quad \dots(1)$$

Differentiating eq. (1) with respect to t

$$\frac{\partial \psi(x, t)}{\partial t} = (-i\omega) Ae^{-i(\omega t - kx)}$$

$$\Rightarrow \frac{\partial \psi(x, t)}{\partial t} = (-i \frac{E}{\hbar}) \psi(x, t)$$

$$\Rightarrow E \psi(x, t) = -\frac{\hbar}{i} \frac{\partial \psi(x, t)}{\partial t} = i\hbar \frac{\partial \psi(x, t)}{\partial t}$$

$$\omega = 2\pi\nu = \frac{2\pi\hbar\nu}{h} = \frac{E}{\hbar} \quad \dots(2)$$

$$[\text{as } \omega = \frac{E}{\hbar}]$$

...(7)

Using equation (6) and (7) in equation (5), we obtained

$$E \psi(x, t) = \frac{p^2}{2m} \psi(x, t) + U(x, t) \psi(x, t) \quad \dots(5)$$

....(8)

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + U(x, t) \psi(x, t)$$

Equation (8) is Time-dependent Schrödinger's equation in one – dimensions.

Three dimensional -Schrödinger's equation

(Time dependent case) :- we know One Dimensional Schrödinger eqn $i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + U(x,t)\psi(x,t)$

In three dimensions it shall be

$$i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m} \left[\frac{\partial^2 \psi(\vec{r},t)}{\partial x^2} + \frac{\partial^2 \psi(\vec{r},t)}{\partial y^2} + \frac{\partial^2 \psi(\vec{r},t)}{\partial z^2} \right] + U(\vec{r},t)\psi(\vec{r},t)$$

$$i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r},t) + U(\vec{r},t)\psi(\vec{r},t) \quad \text{.....(9)}$$

For free particle where $U(\vec{r},t) = 0$

$$i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m} \left[\frac{\partial^2 \psi(\vec{r},t)}{\partial x^2} + \frac{\partial^2 \psi(\vec{r},t)}{\partial y^2} + \frac{\partial^2 \psi(\vec{r},t)}{\partial z^2} \right]$$

$$i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r},t) \quad \text{....(10)}$$

Time independent Schrödinger's equation

In most of the physical system the potential energy does not depend on the time but depends only on the position of the particle, *i.e.*

$U(\vec{r}, t) = U(\vec{r})$ For such a system, the wave function $\psi(\vec{r}, t)$ can be separated into time independent and time dependent part as

$$\psi(\vec{r}, t) = Ae^{-i\omega t} e^{i\vec{k} \cdot \vec{r}} = \psi(\vec{r}) e^{-i\omega t} = \psi(\vec{r}) e^{-\frac{iE}{\hbar} t}$$
$$\frac{\partial \psi(\vec{r}, t)}{\partial t} = -\frac{iE}{\hbar} \psi(\vec{r}) e^{-\frac{iE}{\hbar} t} \quad \dots(11)$$

From time-dependent Schrödinger's 3D equation

$$i\hbar \frac{\partial \psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}, t) + U(\vec{r}, t) \psi(\vec{r}, t) \quad \dots(12)$$

Schrödinger's equation

Time independent case

From equations (11) and (12), we have

$$\begin{aligned} i\hbar\left[-\frac{iE}{\hbar}\psi(\vec{r})e^{-\frac{iE}{\hbar}t}\right] &= -\frac{\hbar^2}{2m}\nabla^2[\psi(\vec{r})e^{-\frac{iE}{\hbar}t}] + U(\vec{r})\psi(\vec{r})e^{-\frac{iE}{\hbar}t} \\ \Rightarrow e^{-\frac{iE}{\hbar}t}E\psi(\vec{r}) &= -\frac{\hbar^2}{2m}e^{-\frac{iE}{\hbar}t}\nabla^2\psi(\vec{r}) + U(\vec{r})\psi(\vec{r})e^{-\frac{iE}{\hbar}t} \\ \boxed{E\psi(\vec{r})} &= -\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r}) + U(\vec{r})\psi(\vec{r}) \end{aligned} \quad \text{....(13)}$$

The above equation is called time independent Schrödinger equation. For the free particle, $U(\vec{r}) = 0$ the time independent Schrödinger equation becomes

$$\boxed{E\psi(\vec{r}) = -\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r})} \quad \text{....(14)}$$

Schrödinger's equation

Time independent case

Equation (13) can also be written as

$$E \psi(\vec{r}) = \left[-\frac{\hbar^2}{2m} \nabla^2 + U(\vec{r}) \right] \psi(\vec{r})$$
$$E \psi(\vec{r}) = H \psi(\vec{r}) \quad \dots(15)$$

As seen from equation (15), H and E behave as operators

$$H = \left(-\frac{\hbar^2}{2m} \nabla^2 + U(\vec{r}) \right) \quad \dots(16)$$

$$E = -\frac{\hbar}{i} \frac{\partial}{\partial t}$$

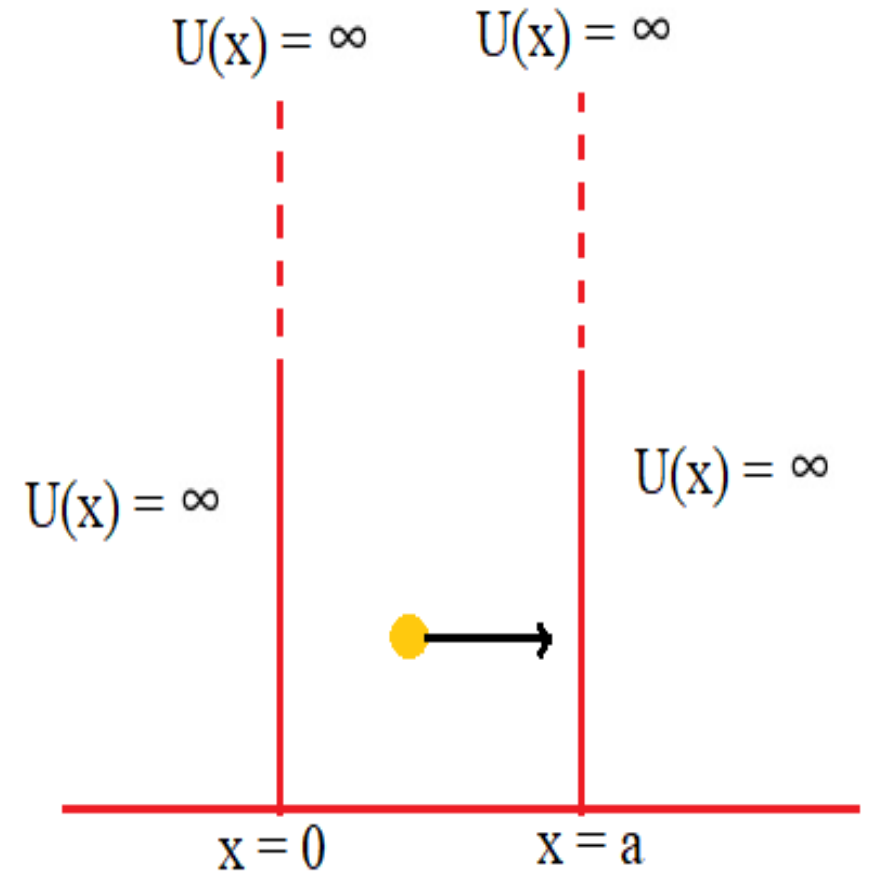
H, the total energy (KE + PE) operator, is called Hamiltonian operator but it depends on position only, and E is called energy operator, which depends on time only.

Free Particle in One Dimensional Potential Box

- Let us consider the motion of a particle of mass m along x -axis in one dimensional potential box which has perfectly rigid walls of infinite potentials at $x = 0$ and $x = a$.
- Thus the motion of particle is within $x = 0$ and $x = a$ between which the potential $U(x) = 0$. Mathematically the potential energy of this particle can be written as

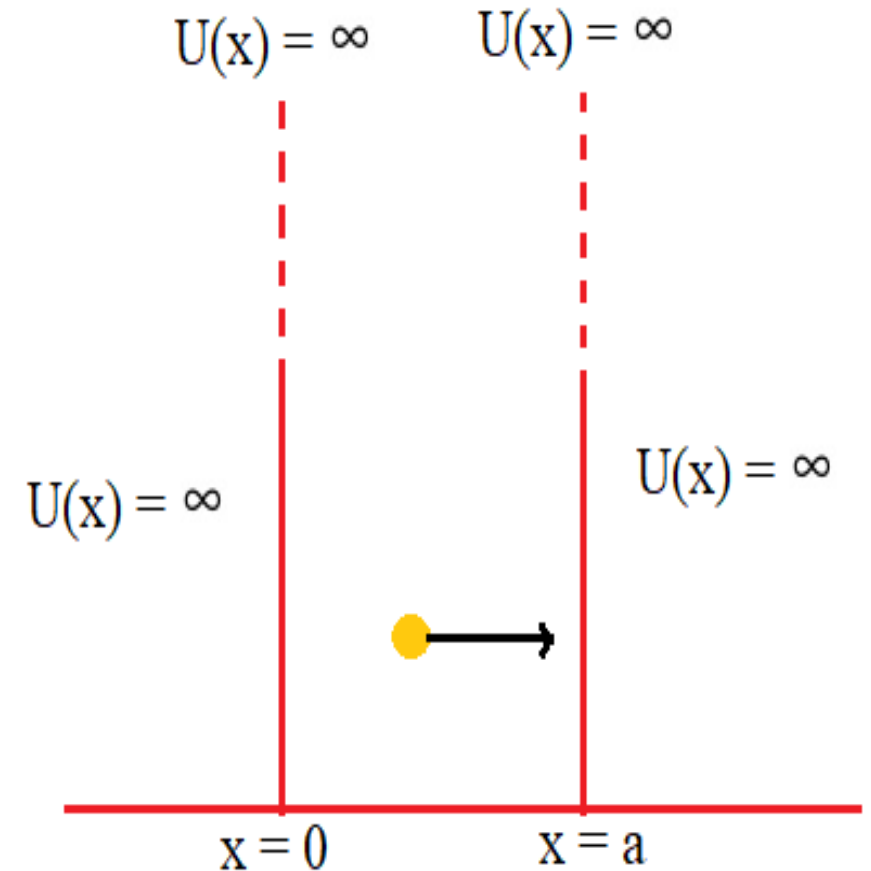
$$U(x) = 0 \text{ when } 0 < x < a$$

$$U(x) = \infty \text{ when } x \leq 0 \text{ and } x \geq a$$



Free Particle in One Dimensional Potential Box

- Hence the particle moves as free in the region $0 < x < a$, but when it reaches at $x = 0$ or $x = a$, a force $-\frac{dU(x)}{dx} = \infty$ acts on the particle due to infinite potential, opposite to the motion.
- As a result the direction of particle changes. Hence the particle remains in the region $0 < x < a$ and not in the regions $x \leq 0$ and $x \geq a$.
- Thus the wave function $\psi(x)$ associated with the particle should be zero outside the box i.e., $\psi(x) = 0$ when $x \leq 0$ and $x \geq a$.

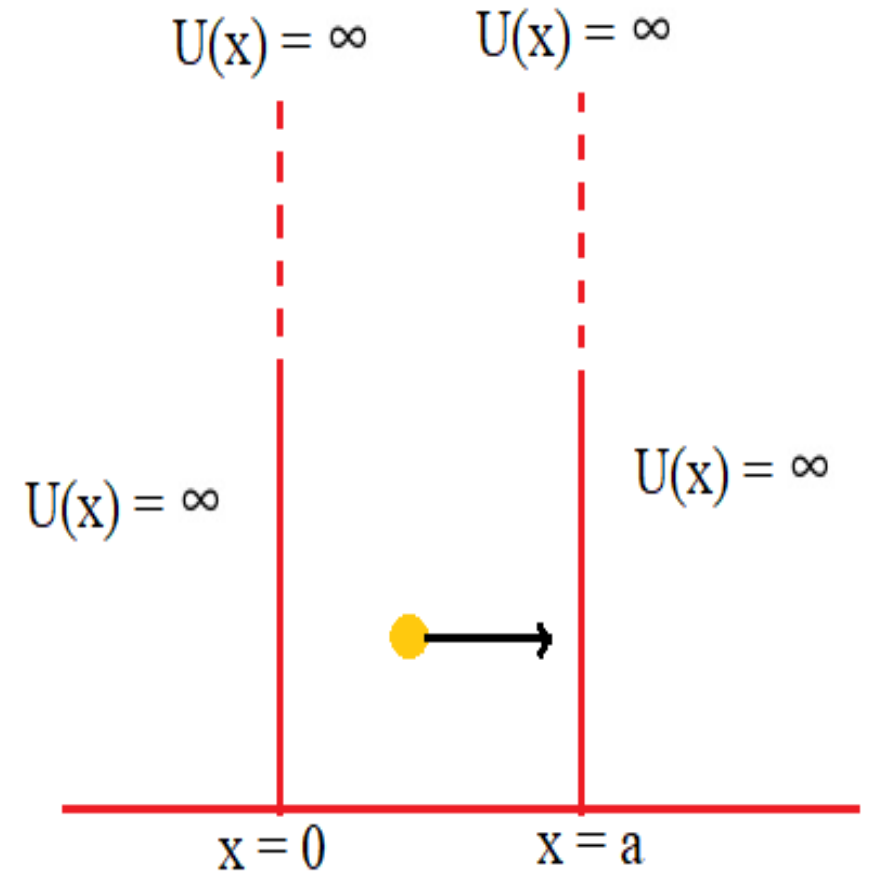


Free Particle in One Dimensional Potential Box

➤ Now in the region $0 < x < a$, the potential is zero, so the time independent Schrödinger equation describing the quantum mechanical motion of particle can be written as

$$E \psi(x) = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2}$$
$$\Rightarrow \frac{\partial^2 \psi(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \psi(x) = 0 \quad \dots(1)$$

where E is the total energy of the particle.
From equation (1)



Free Particle in One Dimensional Potential Box

- From equation (1)

$$\frac{\partial^2 \psi(x)}{\partial x^2} + k^2 \psi(x) = 0 \quad \dots(2)$$

where $k^2 = \frac{2mE}{\hbar^2}$

The general solution of this equation is given by

$$\psi(x) = A \sin kx + B \cos kx \quad \dots(3)$$

where A and B are constants which can be determined by applying boundary conditions.

Constants A and B: For a given one-dimensional box, $U(x) = \infty$ at $x = 0$ and $x = a$. Thus the wave function $\psi(x)$ should be zero at these boundaries.

$$\psi(x) = 0 = A \sin k \cdot 0 + B \cos k \cdot 0$$

$$\Rightarrow B = 0$$

Free Particle in One Dimensional Potential Box

Thus the wave functions of a particle moving in one dimensional box is,

$$\psi(x) = A \sin kx \quad \dots(4)$$

To determine the constant A, we use normalization condition of wave function.

$$\int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx = 1$$

$$\int_0^a A^2 \sin^2 kx dx = 1$$

$$A^2 \frac{a}{2} = 1$$

$$A = \sqrt{\frac{2}{a}} \quad \dots(5)$$

From equations (4) and (5), wave function of a particle moving in one dimensional box

$$\psi(x) = \sqrt{\frac{2}{a}} \sin kx \quad \dots(6)$$

Free Particle in One Dimensional Potential Box

Eigen functions and Eigen values: The solutions of equation (1) are called eigen functions. The eigen functions are

$$\psi(x) = \sqrt{\frac{2}{a}} \sin kx$$
$$0 = \sqrt{\frac{2}{a}} \sin ka$$

$$\Rightarrow ka = n\pi$$

...(7)

where $n = 1, 2, 3 \dots$

[Note: If $n=0$ it means that $\psi(x) = 0$, i.e., probability of finding particle inside the box is zero, whereas the particle is in the box. Hence n cannot be zero.]

Thus the eigen function for a free particle trapped in one dimensional potential box is given as

$$\psi(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$



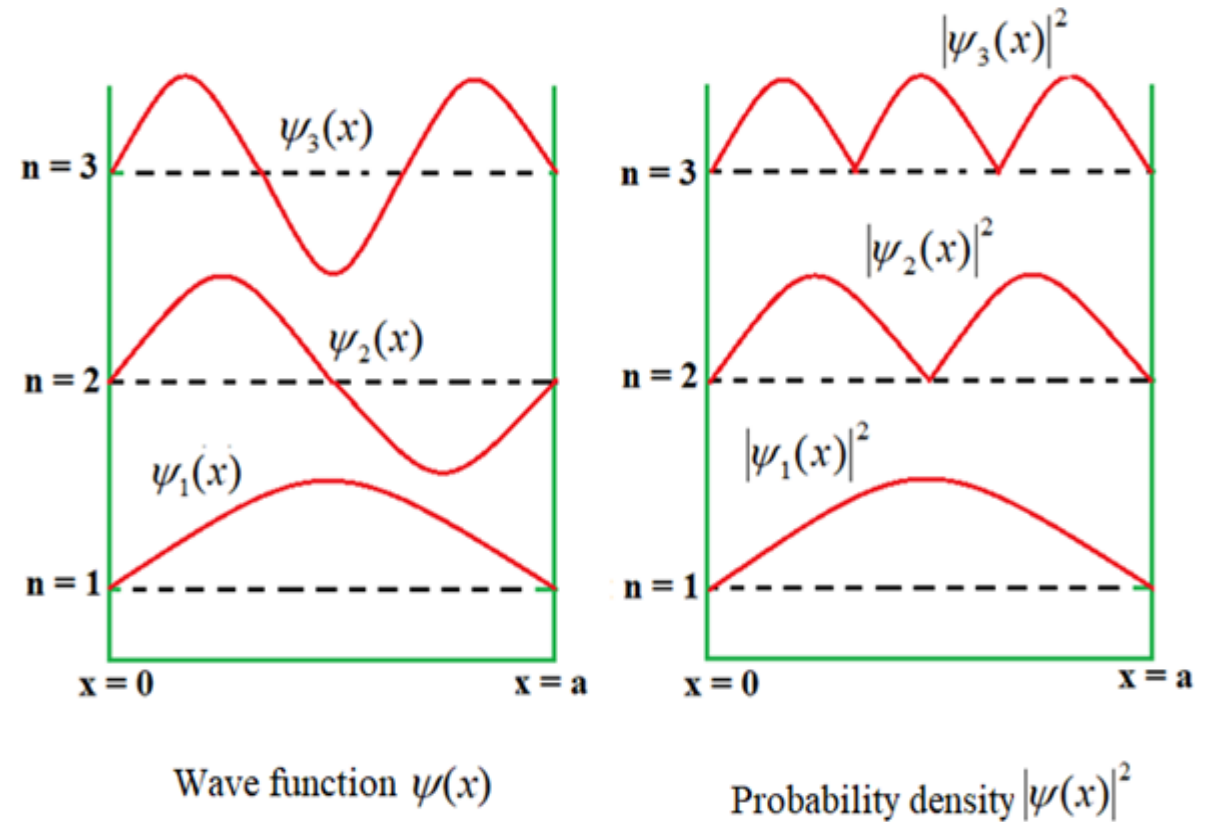
Eigen Function

...(8)

Free Particle in One Dimensional Potential Box

Eigen functions ψ_1 , ψ_2 , and ψ_3 and probabilities $|\psi_1|^2$, $|\psi_2|^2$ and $|\psi_3|^2$ for $n = 1, 2, 3$, are shown in figure as a function of x .

$$\psi(x) = \sqrt{\frac{2}{a}} \sin kx$$

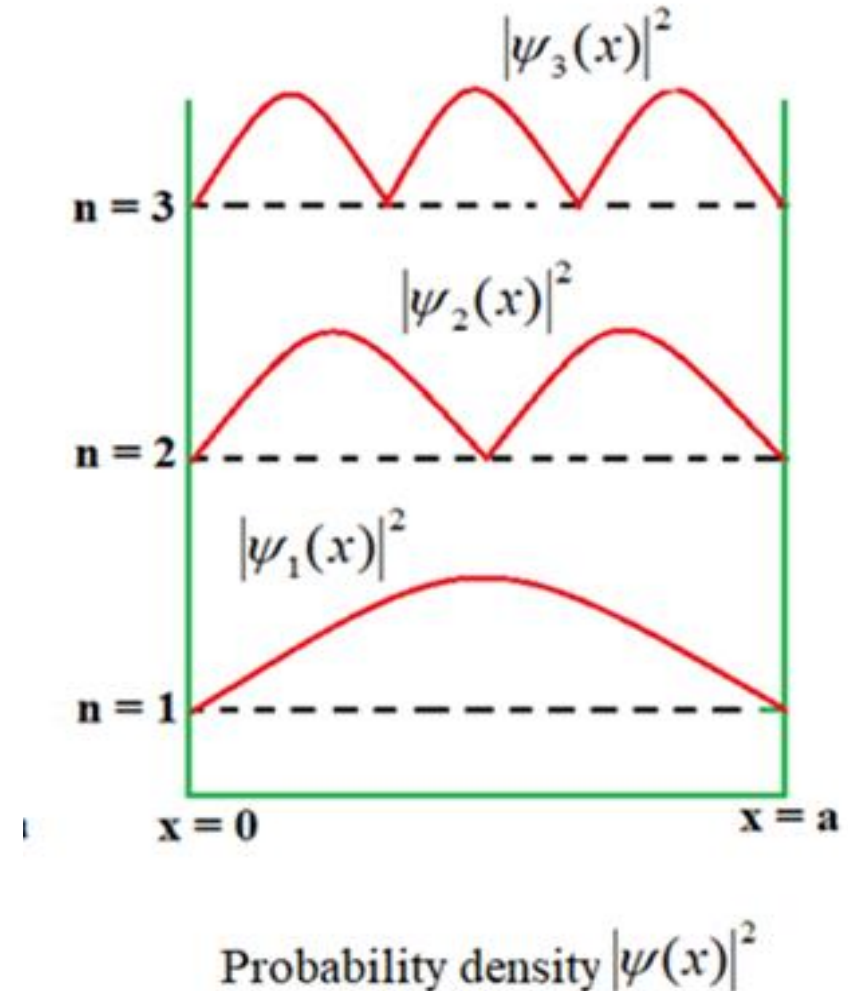


It is evident that nodes are formed at the potential wall of the box and the number of nodes in the range $0 < x < a$ is $(n - 1)$.

Free Particle in One Dimensional Potential Box

Classical Physics: The particle can exist anywhere in the box and follow a path in accordance to Newton's Laws.

Quantum Physics: The particle is expressed by a wave function and there are certain areas more likely to contain the particle within the box.



Free Particle in One Dimensional Potential Box

The wavelengths associated with the matter wave of the particle inside one-dimensional box are given by

$$\lambda = \frac{2\pi}{k} = \frac{2\pi a}{n\pi}$$
$$\lambda = \frac{2a}{n} \quad \dots(9)$$

So, the size of the box will be

$$a = \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2} \dots \dots \dots \quad \dots(10)$$

Now from equation (7), we have

$$ka = n\pi$$

$$k = \sqrt{\frac{2mE}{\hbar^2}} = \frac{n\pi}{a}$$

$$\Rightarrow E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \quad \dots(11)$$

Eigen values of energy of a particle inside an one dimensional potential box.

Free Particle in One Dimensional Potential Box

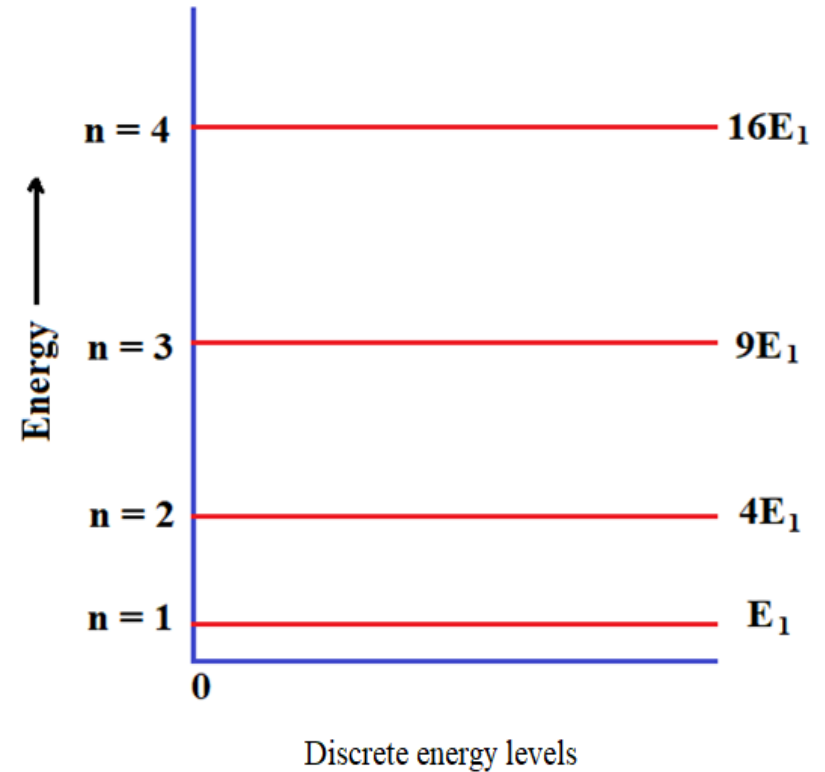
Discrete energy levels:

From equation (11) , $E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$, $E_2 = \frac{4\pi^2 \hbar^2}{2ma^2}$, $E_3 = \frac{9\pi^2 \hbar^2}{2ma^2}$, **...(12)**

Hence particle E_1 : E_2 : E_3 ... = 1:4:9:.....

It is clear from equation (12) that when a particle is limited in a box, the particle cannot have arbitrary energy. The particle can have only specific values or eigen values of energy due to boundary conditions applied on the wave functions at the boundary of the box. These eigen values depends on the mass of the particle m , the size of the box a and the quantum number n .

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$



Thus according to it, the energy of the particle bounded within a box is quantized.

Free Particle in One Dimensional Potential Box

One can draw the following conclusions:

➤ The lowest energy of the particle is obtained by putting $n = 1$. One obtains for $n = 1$ as

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$$

It is called the ground state energy level of the particle. The value of E_n in terms of E_1 are given by

$$E_n = n^2 E_1$$

Where n = quantum numbers

E = Energy of energy levels.

The quantum states with higher values of n ($=2,3,4,\dots$) are termed as excited states.

Free Particle in One Dimensional Potential Box

- Each energy eigen value (corresponding to $n = 1, 2, 3, 4, \dots$) has just one eigen function (corresponding to that integer n). So such energy levels are known as “non-degenerate” energy level.
- When a particle motion is limited inside one dimensional potential box, the particle cannot have arbitrary energy but have only specific values or eigen values. These eigen values depends on the mass of the particle ‘ m ’, the size of the box ‘ a ’ and the quantum numbers ‘ n ’.

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2} \quad E_2 = \frac{4\pi^2 \hbar^2}{2ma^2} \quad E_3 = \frac{9\pi^2 \hbar^2}{2ma^2} \quad \dots\dots\dots$$

Free Particle in One Dimensional Potential Box

- Another conclusion for the motion of the particle inside one dimensional box can also be drawn that the particle cannot have zero energy i.e. the particle cannot be at rest.

[If one assumes particle is at rest i.e. $v = 0$ inside the potential box, the de Broglie wave associated with it will be $\lambda = (h/mv) = \infty$. This is absurd result because there should be the node of wave function at the boundary.]

- The dependence of E on n^2 results in the increased spacing between energy levels as n increases

$$\Delta E = E_{n+1} - E_n = [(n+1)^2 - n^2]E_1$$

- Since the energy E_1 is proportional to $(1/a^2)$, the smaller the box, the greater are the energy spacings (ΔE), i.e., the quantization is more pronounced. Alternatively, as the box is made wider ΔE decreases and in the limited case of a ∞ (free particle) the quantization vanishes.

**Thank
You**

